

DNQ-DLC: dynamic physical graph, bounded tensor interface, quality-guided latent planning

a Runtime graph

Map-relative telemetry
All surrounding vehicles scored
K=3 masked relation tensor
Vehicle identities may change



rebuild at every real step

b Graph-DLC model

Self/relation encoders
Masked invariant aggregation
Recurrent latent transition
State, reward, risk, uncertainty

c Candidate rollouts

Rule anchor + learned proposal
Handcrafted racing maneuvers
H-step latent rollouts
Slots fixed only inside each rollout

d Quality decision

Progress and overtaking
Track quality and proximity risk
Uncertainty penalty
Execute first action + recovery guard

execute

observe $t+1$

rebuild graph

replan